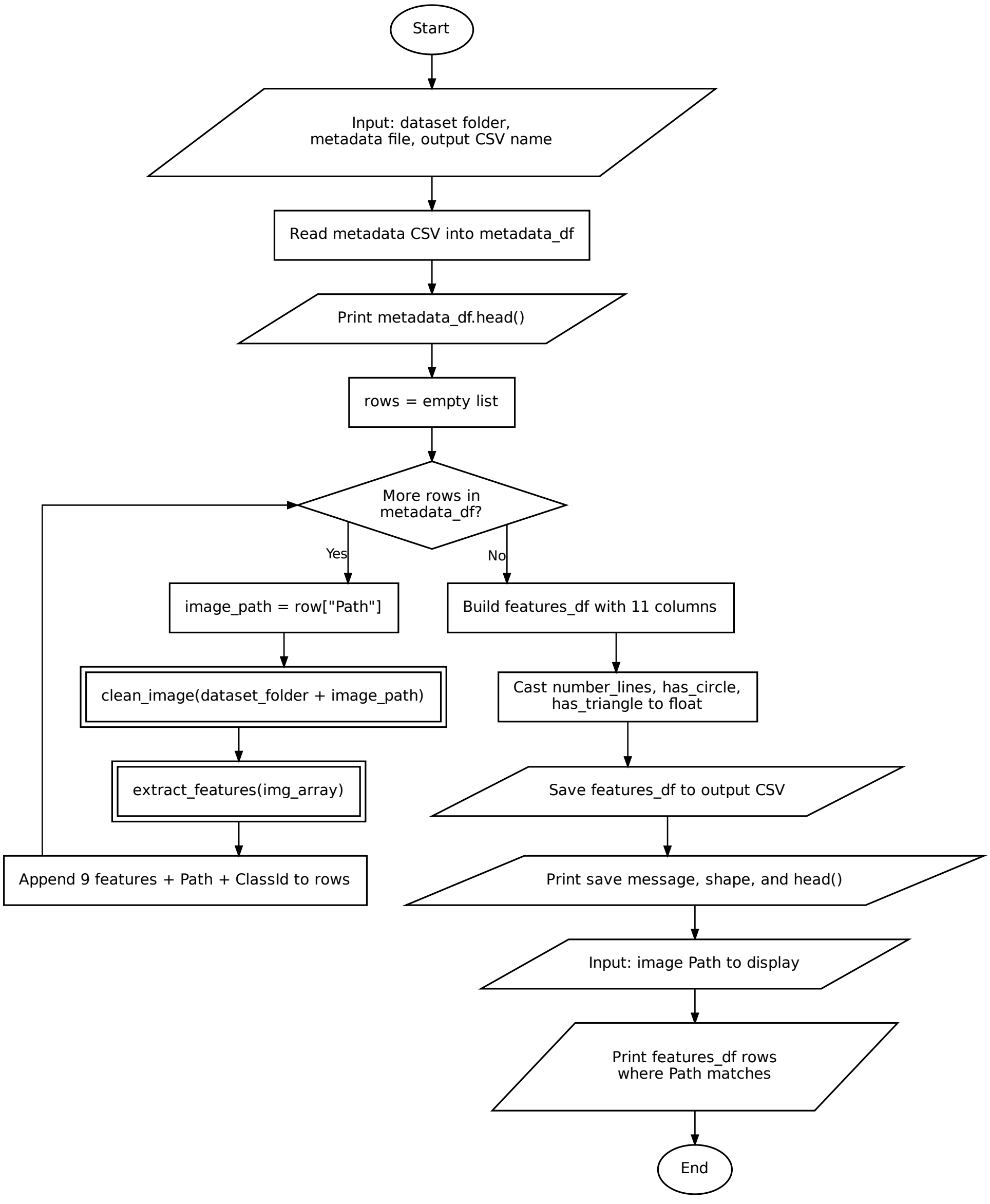
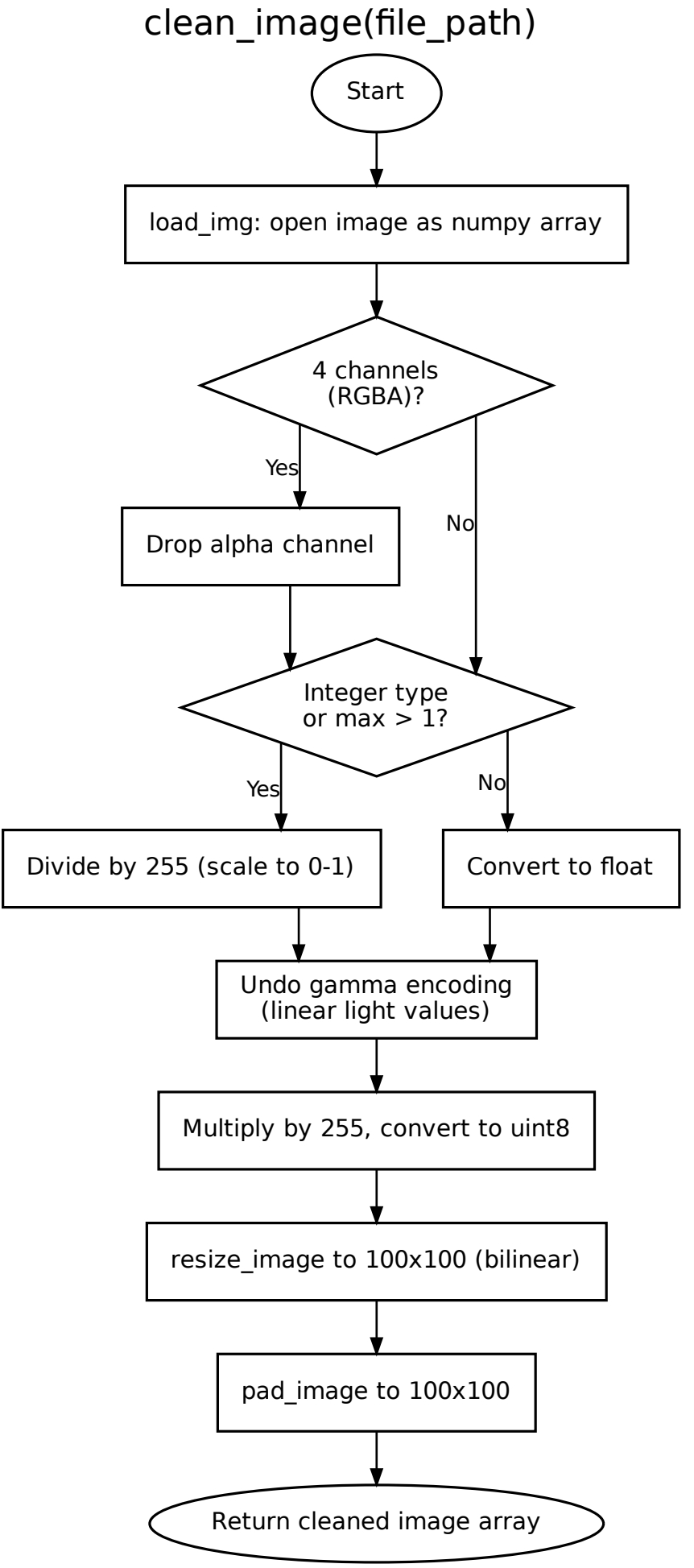
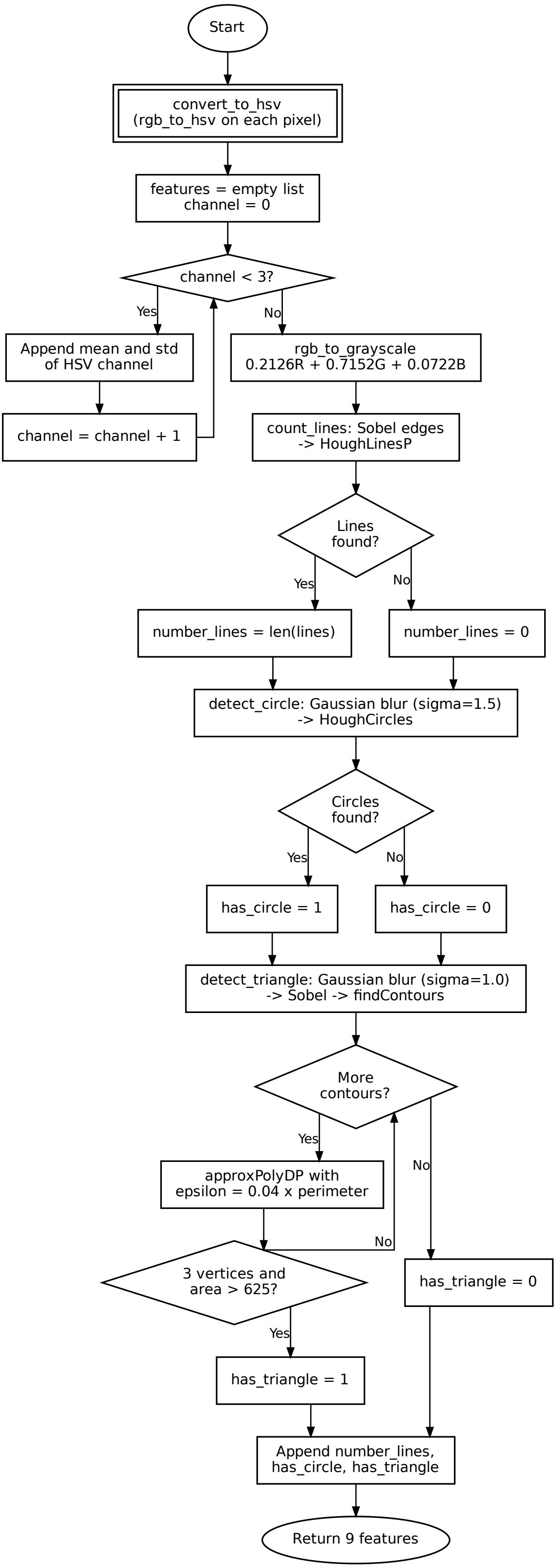
main()


Start
Input: dataset folder,
metadata file, output CSV name
Read metadata CSV into metadata_df
Print metadata_df.head()
rows = empty list
More rows in
metadata_df?
Yes
No
image_path = row["Path"]
clean_image(dataset_folder + image_path)
extract_features(img_array)
Append 9 features + Path + ClassId to rows
Build features_df with 11 columns
Cast number_lines, has_circle,
has_triangle to float
Save features_df to output CSV
Print save message, shape, and head()
Input: image Path to display
Print features_df rows
where Path matches
End


clean_image(file_path)
Start
load_img: open image as numpy array
4 channels (RGBA)?
Yes
No
Drop alpha channel
Integer type or max > 1?
Yes
No
Divide by 255 (scale to 0-1)
Convert to float
Undo gamma encoding (linear light values)
Multiply by 255, convert to uint8
resize_image to 100x100 (bilinear)
pad_image to 100x100
Return cleaned image array

extract_features(img_array)